

QUESTION PAPER - SET 4

SECTION 1 - PERSONAL DATA

1. Name

2. Mobile Number

3. Class 10 Overall Percentage

4. Class 12 PCM Percentage

5. What is your percentage in Pre Sea Training till 4th Semester?

6. Tell us about your strengths.

7. Tell us about your weaknesses.

8. Have you participated in any extracurricular activities during your Pre Sea Training? If yes, share details.

9. Tell us about your family.

10. Where do you see yourself after 5 years?

SECTION 2 - COMPREHENSION PASSAGE

In the late 21st century, the Global Council enacted the "Resource Equilibrium Accord" to address escalating conflicts over dwindling freshwater reserves. The Accord introduced a tiered quota system, assigning water access rights based on each nation's ecological footprint, technological capability for desalination, and per-capita consumption history.

While wealthier nations invested in advanced desalination plants and atmospheric water harvesters, poorer nations — despite having smaller ecological footprints — struggled to meet their citizens' basic needs under their reduced quotas.

Critics argued that the Accord unintentionally entrenched inequality, as historical consumption patterns favored nations that had previously overused water.

To address the backlash, an amendment proposed the creation of a "Water Equity Fund" financed by a surcharge on nations exceeding their quota. The fund would support sustainable water projects in under-resourced regions.

Yet, debates continued over enforcement: how to monitor usage accurately, prevent data manipulation, and ensure that the fund genuinely benefited those in need.

1. What was the main goal behind the introduction of the Resource Equilibrium Accord in the late 21st century?

- A. To establish a global water trading market
- B. To balance global water distribution and prevent conflicts over freshwater resources
- C. To reward nations with low per-capita water consumption
- D. To increase funding for desalination research

2. Which aspect most contributed to the Resource Equilibrium Accord unintentionally reinforcing inequality among nations?

- A. Technological advancements in desalination
- B. Inaccurate monitoring of water usage
- C. Historical consumption patterns that benefited countries with prior high water use
- D. Opposition to the Water Equity Fund

3. Which conclusion is best supported by the information provided in the passage about the Resource Equilibrium Accord?

- A. The amendment eliminated all concerns about fairness
- B. Nations with smaller ecological footprints were given larger quotas
- C. Wealthier nations opposed atmospheric water harvesting
- D. The Accord unintentionally reinforced pre-existing inequalities

4. Under the Resource Equilibrium Accord, when would a nation be required to pay a surcharge to support the Water Equity Fund?

- A. When its per-capita consumption is below the global average

- B. When it fails to reduce its ecological footprint
- C. When it does not participate in atmospheric water harvesting
- D. When its water usage exceeds the assigned quota

5. What is the main challenge that remains unresolved regarding the amendment to the Resource Equilibrium Accord?

- A. Lack of sustainable water project ideas
- B. The unwillingness of wealthier nations to pay surcharges
- C. Uncertainty in the fund's revenue source
- D. Ensuring transparency and preventing misuse of funds

SECTION 3 - COMPREHENSION PASSAGE

The Sundarbans, located in the delta region of India and Bangladesh, is the largest mangrove forest in the world. It is home to diverse wildlife, including the famous Royal Bengal Tiger. The mangroves act as a natural barrier, protecting coastal areas from cyclones and storm surges. They also provide livelihoods for local communities through fishing, honey collection, and eco-tourism. However, climate change poses a serious threat to the Sundarbans. Rising sea levels, increasing salinity, and frequent storms are causing habitat loss.

Conservation projects are being implemented, focusing on planting more mangroves, reducing illegal logging, and promoting sustainable fishing practices. International cooperation is considered vital for the long-term preservation of this unique ecosystem.

1. Which of the following is NOT listed as a livelihood activity for local communities in the Sundarbans, according to the passage? (1 Point) Select the option that is not mentioned as a source of livelihood.

- A. Agriculture
- B. Honey collection
- C. Fishing
- D. Eco-tourism

2. Which of the following is a direct impact of climate change on the Sundarbans ecosystem? (1 Point) Select the option that best represents a climate change effect described in the passage.

- A. Increasing salinity of water
- B. Introduction of new wildlife species
- C. Growth in honey collection
- D. Expansion of fishing zones

3. What is one of the key protective roles played by the mangrove forests in the Sundarbans? (1 Point) Select the option that best describes how mangroves benefit the coastal environment.

- A. They increase the frequency of tidal waves.
- B. They are a major source of underground minerals.
- C. They supply drinking water to nearby villages.
- D. They act as a shield against cyclones and storm surges.

4. What is the primary objective of the conservation efforts described in the Sundarbans passage? (1 Point) Select the option that best reflects the main goal of these projects.

- A. To increase honey production for export
- B. To encourage local communities to move to urban areas
- C. To restore and safeguard the mangrove forests and their biodiversity
- D. To expand commercial fishing operations

5. Why does the passage emphasize the necessity of international cooperation for preserving the Sundarbans? (1 Point) Select the option that best explains the importance of cross-border collaboration.

- A. The Sundarbans ecosystem spans across India and Bangladesh, requiring joint efforts.
- B. Local communities are unwilling to support conservation projects.
- C. The region attracts international tourists who fund conservation.
- D. Climate change is only a concern for a few developed countries.

SECTION 4 - ENGLISH APTITUDE

1. Select the sentence that has the same meaning as: "The cadet received a warning for not adhering to procedures."

- A. The cadet received a warning because he did not adhere to procedures.
- B. The cadet received a warning although he adhered to procedures.
- C. The cadet received a warning so that he would adhere to procedures.
- D. The cadet received a warning even though he adhered to procedures.

2. Identify the correct meaning of the idiom: "To toe the line"

- A. To cross a boundary
- B. To obey rules or authority
- C. To delay action
- D. To take risks

3. She can solve navigation problems as quickly as _____ officer on board.

- A. Many
- B. Every
- C. Any
- D. Some

SECTION 5 - NUMERICAL APTITUDE

1. If \$4,000 is invested at 5% per annum simple interest, how much interest will be earned in 4 years?

- A. \$1,000
- B. \$800
- C. \$700
- D. \$600

2. Choose the correct option in the sequence.

- A. Option A
- B. Option B
- C. Option C
- D. Option D

3. The length of a rectangle is twice its breadth. If its area is 200 square units, what is the breadth?

- A. 5
- B. 8
- C. 10
- D. 20

SECTION 6 - MARITIME KNOWLEDGE

1. Which type of fire extinguisher is best suited for electrical fires?

- A. Water extinguisher
- B. Foam extinguisher
- C. CO₂ extinguisher
- D. Dry sand

2. What does the term “lee shore” refer to in seamanship?

- A. A shore protected from wind
- B. A shore facing the wind
- C. A dangerous shore onto which wind blows
- D. A shallow coastal area

3. Which emergency equipment is used to send distress signals to nearby ships?

- A. EPIRB
- B. Gyro Compass
- C. Echo Sounder
- D. Barograph

4. The observer’s prime vertical passes through the east and west points of which of the following?

- A. Rational horizon
- B. Equinoctial
- C. Celestial meridian
- D. Great circle

5. What is the purpose of ballast water in a ship?

- A. To clean cargo tanks
- B. To improve ship stability and trim

- C. To increase engine efficiency
- D. To cool cargo holds

6. Which type of weather front is most commonly linked to severe thunderstorms, heavy rain, and sudden gusty winds?

- A. Cold front
- B. Occluded front
- C. Warm front
- D. Stationary front

7. What does “abandon ship” mean?

- A. Leaving cargo operations
- B. Evacuating the vessel during emergency
- C. Docking at a port
- D. Entering restricted waters

8. What is the purpose of a ship’s compass?

- A. To measure sea depth
- B. To determine direction
- C. To measure wind speed
- D. To detect underwater objects

9. Which celestial body is commonly used for obtaining a compass error by azimuth during twilight?

- A. Polaris
- B. Venus
- C. The Sun
- D. Mars

10. What is the purpose of a ship’s whistle or horn?

- A. Entertainment
- B. Distress communication only
- C. Navigational sound signaling

D. Engine warning

11. At which location on Earth does the Coriolis effect reach its greatest strength?

- A. At the Equator
- B. At the Tropics
- C. At 30° latitude
- D. At the Poles

12. In ship stability, GM refers to:

- A. Gross Measurement
- B. General Metric
- C. Metacentric Height
- D. Gravity Measurement

13. What is the SI unit used to measure atmospheric pressure?

- A. Knot
- B. Pascal
- C. Joule
- D. Newton

14. In the Northern Hemisphere, what is the typical direction of wind circulation around a low-pressure area?

- A. Anticlockwise and inward
- B. Clockwise and inward
- C. Clockwise and outward
- D. Anticlockwise and outward

15. In navigation, what does “dead reckoning” primarily rely upon?

- A. Radar observations
- B. Celestial sightings
- C. Estimated course and speed
- D. Satellite communication

16. Aneroid barometers are typically calibrated to display atmospheric pressure in which of the following units?

- A. Inches of mercury and millibars
- B. Feet of mercury and millibars
- C. Inches of mercury and millimeters
- D. Inches of mercury and centimeters

17. What is the primary purpose of a ship's freeboard?

- A. To increase cargo capacity
- B. To improve engine efficiency
- C. To provide reserve buoyancy and safety
- D. To reduce fuel consumption

18. What is the function of a ship's rudder?

- A. To increase stability
- B. To change the ship's direction
- C. To reduce rolling
- D. To measure speed

19. What does the term "angle of loll" describe in ship stability?

- A. The angle at which the center of gravity is vertically above the center of buoyancy
- B. The time taken for a vessel to complete one roll cycle
- C. KM is greater than GM
- D. The heel angle at which a vessel with negative GM achieves equilibrium due to hull shape

20. What is the primary purpose of a ship's radar?

- A. Cooking assistance
- B. Detecting objects and navigation
- C. Measuring cargo weight
- D. Monitoring fuel usage

21. What does “starboard” refer to on a ship?

- A. Left side
- B. Front side
- C. Right side
- D. Rear side

22. Which chart projection is most commonly used for marine navigation?

- A. Gnomonic Projection
- B. Mercator Projection
- C. Polar Projection
- D. Cylindrical Equal Area Projection

23. Which gas is primarily responsible for the greenhouse effect?

- A. Oxygen
- B. Nitrogen
- C. Carbon dioxide
- D. Helium

24. Which factor mainly affects ocean tides?

- A. Wind speed
- B. Earthquake activity
- C. Gravitational pull of the Moon and Sun
- D. Atmospheric pressure only

25. When crossing a traffic separation scheme, the vessel should:

- A. Cross at any convenient angle
- B. Cross at right angles to the traffic flow
- C. Alter course to avoid lanes
- D. Follow the traffic flow

26. What is the main purpose of a ship’s anchor?

- A. To increase vessel speed

- B. To hold the vessel in position
- C. To improve fuel efficiency
- D. To reduce rolling motion

27. Which side of a vessel is indicated by a green navigation light?

- A. Port side
- B. Starboard side
- C. Stern
- D. Masthead

28. What is the main function of a life raft?

- A. Cargo storage
- B. Emergency accommodation after abandoning ship
- C. Firefighting
- D. Navigation support

29. A vessel departs Felixstowe, England at 1000 hrs local time (GMT+0) on 2nd March 2014, bound for Kolkata, India, a distance of 10,080 nautical miles. If the vessel maintains a constant speed of 10 knots, what will be the vessel's estimated time of arrival (ETA) at Kolkata in local time (GMT+5.5)?

- A. 13th April 2014, 1000 hrs LT
- B. 13th April 2014, 0430 hrs LT
- C. 13th April 2014, 1530 hrs LT
- D. 13th April 2014, 1200 hrs LT

30. In celestial navigation, what is the First Point of Aries () primarily used to measure?

- A. Measuring True Altitude
- B. Measuring Azimuth
- C. Measuring Right Ascension
- D. Measuring Longitude

31. At what temperature does the air reach saturation, causing water vapor to begin condensing into liquid droplets?

- A. Temperature at which air becomes saturated
- B. Temperature of condensation clouds
- C. Lowest night temperature
- D. Temperature at which rain begins

32. Which instrument is used to measure the salinity of seawater?

- A. Hygrometer
- B. Sextant
- C. Salinometer
- D. Barograph

33. What is the primary navigational purpose of observing the Sun at its meridian passage?

- A. Applying time correction
- B. Finding longitude
- C. Calculating compass error
- D. Determining latitude

34. Which part of a ship experiences the greatest bending moment in heavy seas?

- A. Midship section
- B. Forecastle
- C. Stern peak
- D. Bridge deck

35. According to ship safety regulations, which of the following must be made weathertight as part of the Conditions of Assignment?

- A. Hatchways situated on the freeboard deck
- B. Sounding pipes on deck
- C. Cargo hold ventilators below deck

D. Scuppers above deck

36. What does SOLAS stand for?

- A. Safety of Life at Sea
- B. Security of Life and Ships
- C. Standard Operations for Large Ships
- D. Safety Operations and Life Saving

37. Which instrument is specifically used to measure the apparent wind direction and speed on a moving ship?

- A. Barometer
- B. Psychrometer
- C. Hygrometer
- D. Anemometer

38. The total area beneath the GZ (righting arm) curve from 0° to 40° of heel is a measure of which of the following vessel characteristics?

- A. The vessel's dynamic stability within that heel range
- B. The vessel's metacentric height (GM)
- C. The vessel's KM
- D. The vessel's maximum righting lever

39. What is the full form of GPS?

- A. Global Positioning System
- B. General Port System
- C. Global Port Signal
- D. Geographic Position Source

40. The vertical distance between the upper edge of the deck line and the centre of the load line disc represents the:

- A. Summer Freeboard
- B. Summer Draft
- C. Fresh Water Allowance

D. Winter North Atlantic Freeboard